



ALLEN
CAREER INSTITUTE
KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
26-01-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Consider a large drop of liquid on a glass plate. Contact angle is θ and density of liquid is ρ . Surface tension is S



(A) $H = \sqrt{\frac{2S}{\rho g}}(1 - \sin \theta)$

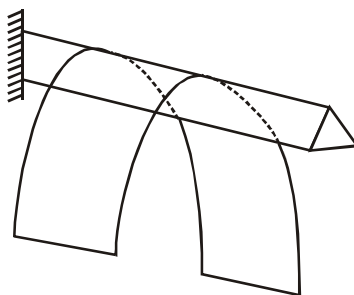
(B) $H = \sqrt{\frac{2S}{\rho g}}(1 + \sin \theta)$

(C) $H = \sqrt{\frac{2S}{\rho g}}(1 + \cos \theta)$

(D) $H = \sqrt{\frac{2S}{\rho g}}(1 - \cos \theta)$

Space for Rough Work

2. A half thin pipe of mass 1 kg and radius 20 cm is balanced on a knife edge. Its time period of small oscillations is :

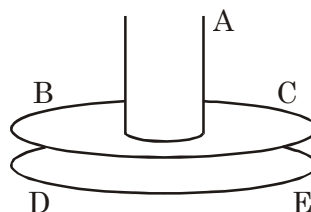
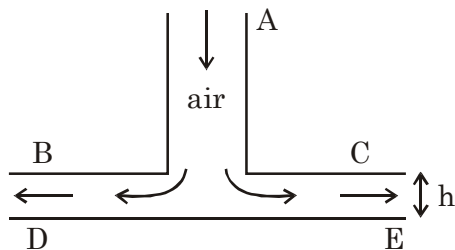


- (A) $\frac{2\pi}{5}$ sec (B) $\frac{\pi}{5}$ sec (C) $2\pi\sqrt{\frac{\left(1 - \frac{2}{\pi}\right)}{10}}$ sec (D) None of these
3. A block of mass 1 kg is kept on a spring of spring constant 100 N/m but is not attached to it. It is pushed down by a distance of 10 cm from its mean position and released. What is the potential energy of the spring after $\frac{\pi}{30}$ sec ?

- (A) 0 (B) $\frac{1}{2}$ J (C) $\frac{1}{8}$ J (D) $\frac{9}{8}$ J

Space for Rough Work

4. DE is a flat horizontal cardboard disc of radius R over which we place another coaxial disc BC of same radius. BC is fitted with a tube A of radius r . On blowing air through A, we find that DE is attracted to disc BC. If the weight of DE is 100π gm, what should be velocity of air through A so that the attractive force can support DE? Density of air as ρ , pressure of air in A is atmospheric. $R = 20$ cm, $r = 5$ cm, $h = 1$ cm, $\rho = 1$ kg/m³. Assume that the flow is laminar.



- (A) 0.5 m/s approx. (B) 13.5 m/s approx.
(C) 18.5 m/s approx. (D) None of these

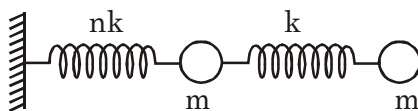
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
-
5. When an object is placed at the proper distance to the left of a converging lens, the image is focused on a screen 30 cm to the right of the lens. A diverging lens is now placed 15 cm to the right of converging lens, and it is found that the screen must be moved 20 cm to the right to obtain a sharp image.
- (A) focal length of diverging lens is -26.25cm.
(B) focal length of converging lens is +20cm.
(C) the object is placed 60 cm to the left of converging lens.
(D) The image formed on screen is inverted in both the cases.
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Space for Rough Work

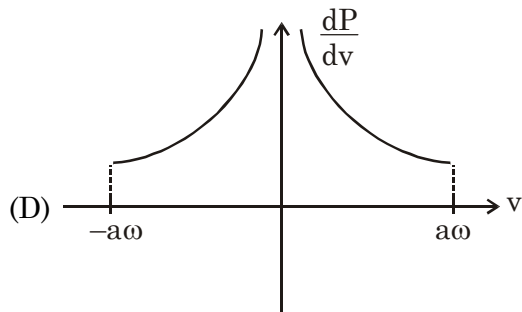
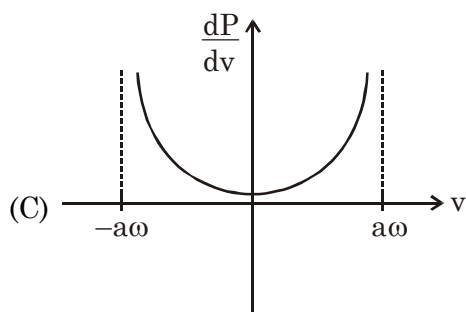
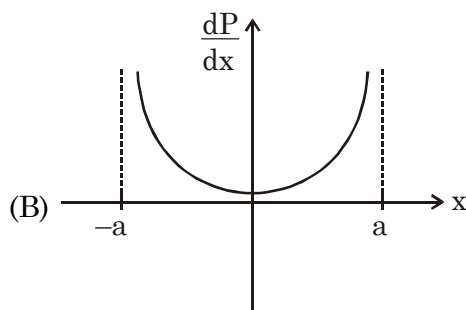
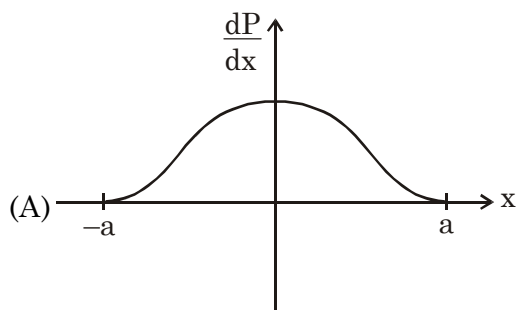
6. Two masses m each lie on a frictionless horizontal table. Spring constants of spring are nk and k respectively (n is a number). The relaxed lengths of both the springs are same. Both the blocks oscillate so that the springs always have same length. Choose **CORRECT** statement(s):



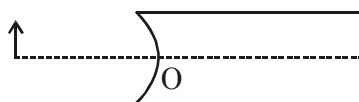
- (A) The blocks oscillate in same phase with same time period.
 (B) $n = 2$
 (C) Amplitude of right mass is double that of left mass
 (D) The given situation is not possible.
7. A spherical globe of radius $r = 10$ cm is kept over a circular plane mirror such that south pole of the globe is at it's center. A man of eye height 1.5 m is viewing the image of the globe in the mirror.
- (A) The radius of mirror should be at least 10 cm for the man to view the equator.
 (B) For the man to view the image of latitude of 37° North, mirror should have at least radius of 12.5 cm.
 (C) For case in option (B), man has to stand at a distance of 132.5 cm from South pole.
 (D) For man to view the equator image (as in option A), man's eyes have to be vertically above the equator.

Space for Rough Work

8. A body is executing SHM on a straight line (x-axis) with time period 2 sec and amplitude 1cm. Assume that probability of the body to be from $x = -1\text{cm}$ to $x = +1\text{cm}$ is 1. Probability of finding particle within interval x to $x + dx$ is dP , probability of particle having velocity from v to $v + dv$ is dP . Find $\frac{dP}{dx}$ and $\frac{dP}{dv}$. Plot them against x & v .



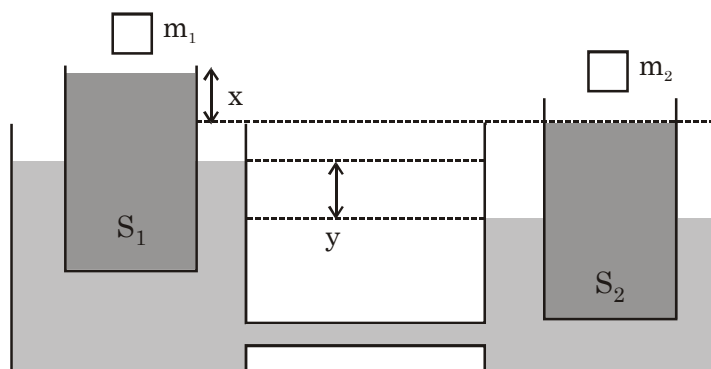
9. A hemisphere of radius 4 cm is carved out of a glass rod of diameter 8 cm & refractive index 1.60. An object 1 mm tall forms an erect image 4 mm tall.



- (A) The image must be formed in air.
 (B) The image is formed at a distance of 32 cm from point O.
 (C) The object is 1 cm away from centre of curvature of hemisphere.
 (D) The image must be real.

Space for Rough Work

10. Two identical cylinders are connected by a pipe. In both cylinders we fill water partially and float two smaller cylinders as shown. The left cylinder has area S_1 & right one has area S_2 . These cylinders are both massless. They are filled with equal volume of water (V). Now in left cylinder we float a block of mass m_1 and in right cylinder we float a block of mass m_2 . Before floating, the level in floating cylinders differed by x_1 & level in lower cylinders differed by y_1 . The same values after floating are x_2 & y_2 respectively.



(A) $x_1 = v \left(\frac{1}{S_1} - \frac{1}{S_2} \right)$

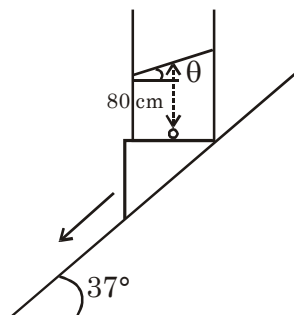
(B) $y_1 = 0$

(C) $x_2 = \frac{(m_1 - m_2)(S_1 - S_2)}{\rho S_1 S_2}$

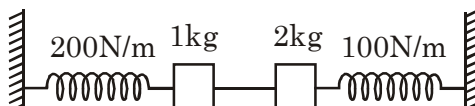
(D) $y_2 = \frac{m_1 - m_2}{\rho S_2}$

Space for Rough Work

11. A cylindrical vessel of radius 1m filled with water is fitted on a platform whose upper surface is horizontal. The platform slides frictionlessly on an inclined plane as shown. There is a small hole in the cylinder at a depth of 80 cm from the top. The vessel is accelerating in steady state.



- (A) The angle made by free surface with horizontal is $\theta = 37^\circ$.
 (B) The water comes out with a speed of 4 m/s relative to the container at this moment.
 (C) The water comes out with a speed less than 4 m/s relative to the container at this moment.
 (D) The water comes out with a speed more than 4 m/s relative to the container at this moment.
12. Two blocks connected by an inextensible string and connected to two springs in relaxed state are kept on a smooth floor as shown. The 2 kg block is pushed by a distance of 1 cm against right spring and released.



- (A) The maximum compression in left spring is $\frac{1}{\sqrt{2}}$ cm .
 (B) The time period of oscillation is $\frac{\pi}{5}$ sec .
 (C) The string is always taut (except at mean position)
 (D) When right block first comes to rest, left spring is at natural length.

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

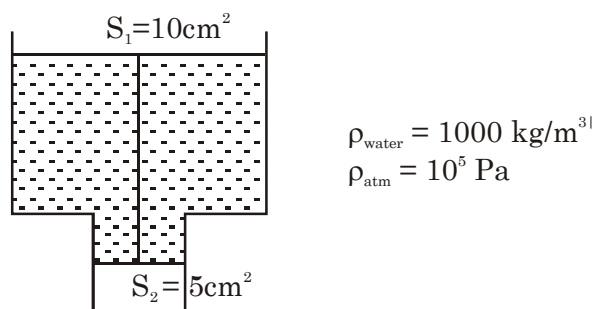
-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

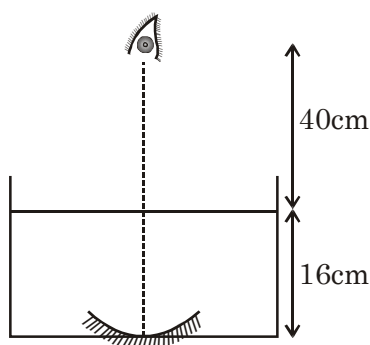
1. Assume that atmosphere above earth extends upto a height of 100 km. Atmosphere has refractive index of 1.01 and earth's radius is 6400km. Find the angle of deviation (in radian) suffered by the Sun rays at time an observer on earth sees sunrise.

Space for Rough Work

2. A vertical cylindrical tube of varying cross section contains water contained between two massless pistons. Find the tension (in N) in the inextensible thread connecting the two pistons. Thread has length of 1m.

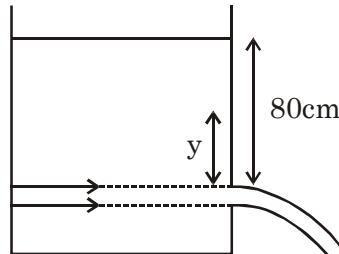


3. A man sees his own image in a concave mirror submerged in a liquid oil tank. The refractive index of oil is 1.6. The depth of oil is 16 cm. The eye sees a real image at a distance of 30 cm from itself. The focal length of the mirror (in cm) in Cartesian coordinate system is

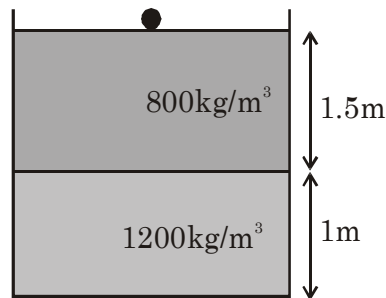


Space for Rough Work

4. A tank of water is full upto 80cm above the hole. The radius of tank is 1m and radius of hole is 1mm. A circular light beam of radius 1mm which covers the hole completely is incident on the hole from inside water. It is seen that the beam is travelling along the water stream like in an optical fibre. At what level of water y (in mm) will the light beam begin to emerge from the curved surface of water stream? $n_{\text{water}} = \frac{4}{3}$

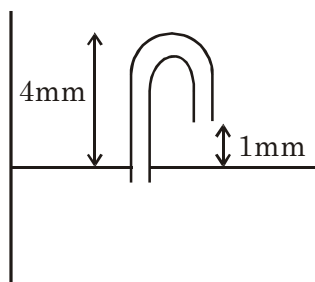


5. A small ball is dropped at the surface of the vessel containing two immiscible liquids as shown. The velocity of ball becomes zero just as it reaches the bottom. What is the density (in kg/m^3) of the balls? Neglect viscosity.



Space for Rough Work

6. A capillary tube of radius 4 mm shaped like an inverted U is lowered into a vessel of water. If the contact angle is 0° , what is the radius of curvature (in mm) of water at free surface. $S = 0.1 \text{ N/m}$, $\rho = 10^3 \text{ kg/m}^3$.



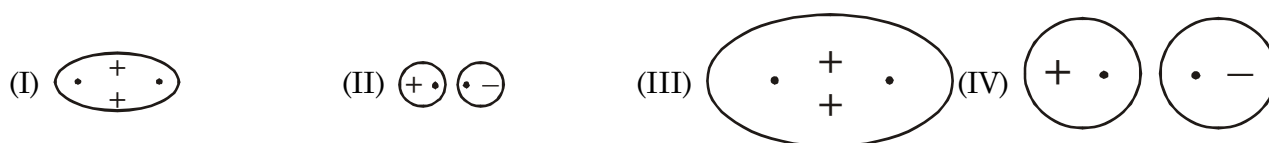
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 12)

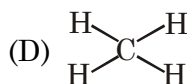
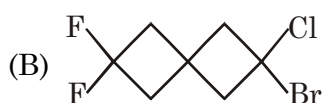
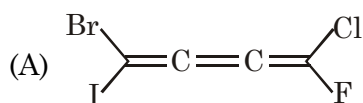
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- In which of the following ionic solid, cation & anion swap will change structure
 (A) NaCl (B) CsCl (C) ZnS (Zinc blende) (D) Na₂O
- A zinc rod weighing 25 g was kept in 100 mL of 1 M CuSO₄ solution. After certain time the molarity of Cu²⁺ in the solution was 0.8 M. The weight of Zn rod left is equal to
 Given : (At. mass of Zn = 65.4)
 (A) 1.308 g (B) 23.692 g (C) 25 g (D) none of these
- Consider LCAO of 1s and 2s orbitals of Li atoms to form Li₂ molecule. Which produce given four molecular orbitals :-



Which of the following statement is/are **CORRECT**?

- M.O. IV is vacant orbital in Li₂
 - Electron pair in M.O. III is responsible for existence of Li₂
 - M.O. II and M.O. IV have common nodal plane
 - Addition and removal of electron take place from M.O. III when Li₂ undergoes in either Li₂⁺ or Li₂⁻ formation
- (A) i, ii, & iv (B) i, ii & iii (C) i & ii only (D) ii & iv only
- Identify optically active compound?

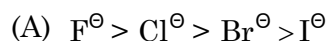


Space for Rough Work

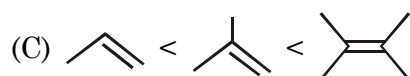
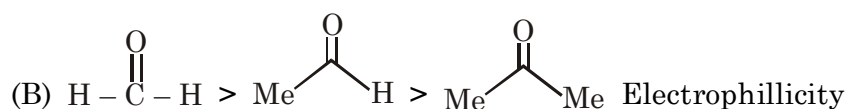
SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
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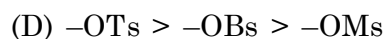
5. Identify correct reactivity order?



Nucleophilicity on DMF



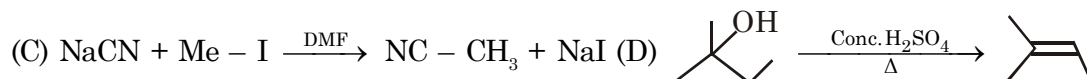
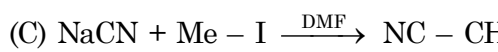
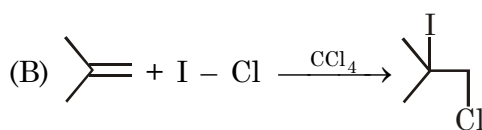
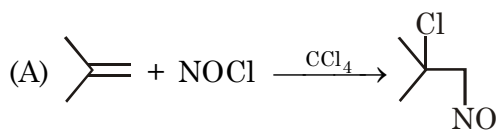
Nucleophilicity order



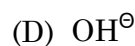
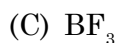
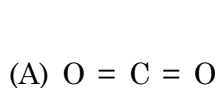
Leaving group ability

Space for Rough Work

6. Identify reactions correctly matched with their major product?



7. Identify total number of electrophiles?



8. Which of the following statements concerning alkaline earth metals is/are **INCORRECT**?

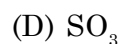
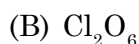
(A) BaF2 is most soluble fluoride among alkaline earth metal fluorides

(B) BeSO3 is least soluble alkaline earth metal sulphites

(C) BeCO3 is most stable carbonate of alkaline earth metal carbonates

(D) Melting point of BeO is lesser as compared to BeF2

9. Which of the following oxide(s) produce atleast one "-ous" and one "-ic" acid on hydrolysis?



Space for Rough Work

10. Select the **CORRECT** statement(s)?
- (A) During the hydrolysis of PCl_5 , POCl_3 will be formed as a intermediate
 - (B) During the hydrolysis, hybridisation of central atom in the transition state of NCl_3 is does not change
 - (C) Hydrolysis of BF_3 produce B(OH)_3 and HF as a final product
 - (D) During the hydrolysis of inter halogen compounds (XX'_n), hydra acid (HX') is formed as a side product (where X and X' are different halogens)
11. Select the correct statement (s) :
- (A) Schottky defect is shown by CsCl
 - (B) Frenkel defect is shown by ZnS
 - (C) hcp and ccp structures have the same coordination number 12
 - (D) On increasing pressure, coordination number of CsCl decreases to that of NaCl
12. Which is **CORRECT** statement(s) -
- (A) Rhombohedral, triclinic and hexagonal crystal system, only primitive Bravais lattice is known
 - (B) In cubic crystal system, $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$
 - (C) In orthorhombic crystal system, $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$
 - (D) In monoclinic crystal system, primitive and face centered bravais lattice are known.

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

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- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

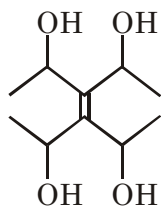
1. C-C bond distance in diamond is $\sqrt{3}\text{\AA}$. What is density of diamond (in gm/cm^3).
 $(N_A = 6 \times 10^{23})$
2. One gram of the chloride of an element was found to have 0.835 gm of chlorine. The vapour density of chloride is 85. If molecular formula of chloride is MCl_x , what is the approximate value of integer x.
3. How many following molecule(s) does/do not exist due to poor shielding of electrons?
 PH_5 , FeI_3 , PbI_4 , BiF_5 , XeF_4 , OF_4 , BiCl_5 , SF_5

Space for Rough Work

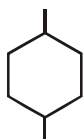
4. Total number of molecules/ions in which L.U.M.O. is π^* among the following will be.

CO, N_2 , O_2 , B_2 , O_2^{2+} , NO^+ , C_2^{2-} , C_2 , CN^-

5. Identify total number of stereoisomers for following compound



6. Calculate total number of monochlorination products for following compound?



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $\tan^{-1}(x+1) + \cot^{-1}(x-1) = \sin^{-1}\frac{4}{5} + \cos^{-1}\frac{3}{5}$, then possible value of x is
 (A) $4\sqrt{\frac{3}{7}}$ (B) $4\sqrt{\frac{7}{3}}$ (C) $-4\sqrt{\frac{7}{3}}$ (D) $6\sqrt{7}$
2. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2020)$ is equal to -
 (A) 3 (B) 4 (C) 5 (D) 6
3. If $\sum_{r=0}^n \frac{(r+5)}{{}^nC_r} = \sum_{r=0}^n \frac{n}{{}^nC_r}$, then the value of 'n' is -
 (A) 6 (B) 10 (C) 14 (D) 18
4. Let $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$, where A_n and B_n are rational numbers and $n \in \mathbb{N}$, then the value of $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$ is equal to -
 (A) $\left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right]$ (B) $\lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]$ (C) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{|\cos x - \sin x|}{\left| x - \frac{\pi}{4} \right|}$ (D) $\left[\lim_{x \rightarrow 0} \frac{\sin \sqrt{2} x}{x} \right]$
 (where $[.]$ denotes greatest integer function)

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Let $f(x) = \begin{cases} ax & ; x < 2 \\ ax^2 - bx + 3 & ; x \geq 2 \end{cases}$ and ' f ' is differentiable $\forall x \in \mathbb{R}$, then -

- (A) $a = \frac{9}{4}$ (B) $a = \frac{3}{4}$ (C) $b = \frac{9}{4}$ (D) $b = \frac{3}{4}$

6. Identify the **CORRECT** statements-

- (A) Sum of two periodic functions may or maynot be a periodic function
 (B) A differentiable function in $\mathbb{R}$ will always be continuous in $\mathbb{R}$.
 (C) A continuous function whose domain is in a closed interval is always bounded
 (D) In $[a, b]$ for a function $f(x)$, if $f(a) \cdot f(b) < 0$, then there exists atleast one root of $f(x) = 0$ in (a, b) .

Space for Rough Work

7. If $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 3}$, then which of the following statement(s) holds good ?

(A) $f(x)$ is continuous at $x = 1$

(B) $f(x)$ is discontinuous at $x = 1$

(C) $\lim_{x \rightarrow 2} f(x)$ exists finitely

(D) $\lim_{x \rightarrow -2} f(x)$ exists finitely

8. If $f(x) = \begin{cases} \cos^{-1}(\cos x) & |x| < \pi \\ \pi \log_{\pi} |x| & |x| \geq \pi \end{cases}$, then correct statement(s) is/are -

(A) $f(x)$ is not differentiable at three points. (B) $f'(\pi^+) + f'(-\pi^-) = 0$

(C) $f'(\pi^+) = f'(-\pi^-)$

(D) $f'(0^+) + f'(0^-) = 0$

9. The function $f(x) = [\cos x] \cos x$ in $(0, 2\pi)$ -

(A) is discontinuous at two points

(B) is non differentiable at two points

(C) has range $(0, 1]$

(D) is such that $f(f(x)) + 1 = 0$ has no real roots.

Space for Rough Work

10. Given $\lim_{x \rightarrow \alpha} \frac{\sin([\![x^3]\!]-[x]^3)}{(x-\alpha)^3}$ exists finitely, then the value of α can be (where $\alpha \in \mathbb{I}$ & $[.]$ denotes

greatest integer function)-

- (A) 0 (B) 1 (C) -1 (D) 2

11. Let $f(x) = \begin{cases} \min\{\sin^{-1} x, \cos^{-1} x\} & ; \quad 0 \leq x \leq 1 \\ \max\{\sin^{-1} x, \cos^{-1} x\} & ; \quad -1 \leq x < 0 \end{cases}$ and $f(x) = \frac{1}{2}$, $f(x) = \frac{1}{4}$ & $f(x) = 2$ has number

of solution(s) $\lambda_1, \lambda_2, \lambda_3$ respectively, then

- (A) $\lambda_1 = 2$ (B) $\lambda_2 = 1$ (C) $\lambda_2 = 2$ (D) $\lambda_3 = 1$

12. For the sequence $\left(\frac{{}^{50}C_0}{1.2.3}\right), \left(-\frac{{}^{50}C_1}{2.3.4}\right), \left(\frac{{}^{50}C_2}{3.4.5}\right), \left(-\frac{{}^{50}C_3}{4.5.6}\right), \dots, \left(\frac{{}^{50}C_{50}}{51.52.53}\right)$ numerically greatest

term is/are-

- (A) 21st term (B) 22nd term (C) 24th term (D) 25th term

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

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●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4\}$. The total number of functions $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ and range of $f(x)$ contains exactly three elements, is
- The number of values of x , $x \in [-2, 3]$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous, is
(where $[.]$ denotes greatest integer function)-
- If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$ where $x, y, z > 0$ then minimum possible value of $(x + y + z)$ is

Space for Rough Work

4. Number of points where $f(x) = |(2^x - 1).(3^x - 3)x.(x - 1).(x - 2)|$ is not differentiable
5. If N denotes the number of positive integral solutions of equation $xy = 2^3 \cdot 3^4 \cdot 5^6.(x + y)$, then N is equal to
6. Let $S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$ then $\cot\left(\lim_{n \rightarrow \infty} S_n\right)$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	12
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator